BINGQI HUANG

Robotics Research Lead | Full-Stack Robotics Engineer

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PROFILE

Robotics researcher and full-stack engineer leading laboratory robotics projects across hardware, ROS software, learning pipelines, and real-world deployment; experienced in mentoring students and coordinating external engineering partners.

EDUCATION

Tsinghua University | Ph.D. Student, Aerospace Science and Technology 2025–Present

Tsinghua University | B.S. & B.Eng., Theoretical & Applied Mechanics / Aerospace Engineering 2021–2025

TU Delft, the Netherlands | Exchange Student, Robotics 2023–2024

RESEARCH & ENGINEERING EXPERIENCE

AGIBOT | AI Algorithm Research Intern Jul 2026–Present

- Initiated and independently lead research on world models for embodied AI.

Distributed and Intelligent Space System Laboratory, Tsinghua University | Robotics Research Lead Present

- **Lead laboratory robotics research and development**, including project direction, robot platform integration, software architecture, and troubleshooting.
- **Mentor undergraduate researchers in robotics research**, system debugging, and participation in the LingBot VLA 2.0 Challenge.
- **Coordinate cross-vendor engineering**, covering machining shops, manipulator and actuator suppliers, OEM partners, procurement, technical requirements, and documentation.
- **Develop and maintain DSSL Robotics repositories**, including original ROS packages and adapted open-source robot-learning systems.

LunarBot | Mobile Manipulation Platform 2024–Present

- **Led end-to-end development as the principal engineer**: mechanical and electrical design, 4WIS–4WID base and RealMan RM75 integration, assembly, ROS development, bring-up, and debugging.
- **Built robot learning and deployment infrastructure**, including a data collection pipeline, VLA training and adaptation, and real-robot policy deployment.
- **Developed a QP-based whole-body controller** for coordinated mobile manipulation.

LunarBot-Mini | Dual-Arm Mobile Robot (in development) 2026–Present

- Lead platform design, hardware integration, ROS development, and supplier coordination for a Ranger Mini base with dual PiPER arms and sensor suites.

HONORS & PATENTS

Tsinghua Graduate Comprehensive Scholarship, Second Class (2026); Future Technology Taihu Scholarship (2024); Xingjian College scholarships (2022, 2023); three mobile robot hardware patent applications.

ROBOT LEARNING & AGENTIC AI RESEARCH

CVAC — Camera-Robust Vision-Language-Action Policies | π0.5 / OpenPI · First Author 2026

- **Proposed cross-view action consistency** for flow-matching policies using paired observations of the same state; inference requires only a single RGB image, without camera calibration.
- **Achieved 87.2% ± 0.4% on LIBERO-Plus** (+7.4 pp vs. baseline); improved real-robot success rates under held-out camera viewpoints from 53.3% to 74.4%.

SCVC — Selective Cross-View Consistency | NVIDIA Cosmos Policy · First Author 2026

- **Adapted NVIDIA Cosmos Policy**, a 2-billion-parameter world-action model, using selective consistency regularization on view-invariant action, proprioceptive-state, and value predictions while preserving view-dependent video predictions.
- **Improved held-out orbital-view success by 12.2 pp** over a matched training control; audited wrist-camera dependence in camera-robustness benchmarks.

Thoth — Trustworthy Agentic Lunar Robotics | Agent Harness · Manuscript in Preparation 2026

- **Built a Codex-like robot agent harness** for goal-driven mission planning and adaptive recovery through typed robotic capabilities, with independent effect verification, authorization and auditable execution traces.
- **Evaluated in NVIDIA Isaac Sim / Space Robotics Bench** with ROS 2 robot skills and PLEXIL / EUROPA procedural-planning baselines under unforeseen lunar-base mission scenarios.

ENGINEERING OPERATIONS & INDUSTRY EXPERIENCE

Robotics Lab & GPU Infrastructure | Technical Owner Tsinghua University

- Maintain robotic equipment (UR arms, Robotiq grippers and LunarBots); own repairs, spare parts, and procurement. Built and maintain GPU servers and coordinate vendor support.

Ericsson China | Mechanical Engineering Intern Jul–Sep 2024

- Designed antenna enclosure CAD, ran ANSYS impact and composite simulations, documented workflows, and evaluated suppliers.

SELECTED PUBLICATIONS

B. Huang et al., “Cross-View Action Consistency for Camera-Robust Vision-Language-Action Policies,” arXiv:2608.06965 (2026).

B. Huang et al., “Selective Cross-View Consistency for World Action Models,” arXiv:2608.21402 (2026).

B. Huang et al., “Toward Self-Evolving Lunar Robotic Autonomy Through Contract-Governed Skill Registration,” Astronautics 1(3), 15 (2026).

B. Huang et al., “Whole-Body Control of a Holonomic Wheeled Mobile Manipulator for Coverage-Continuous Surface Interaction on Lunar Infrastructure,” IAC 2026.

C. Wei, B. Huang, et al., “Gesture-Driven Action Generation for Lunar Robotic Systems Based on Multimodal Large Language Models,” IAC 2026.

C. Wei, B. Huang, et al., “MOAL: Mutual Odometry and Localization for Lunar Robot Cluster,” IAC 2026.

TECHNICAL SKILLS

Robotics & Control: ROS / ROS 2, CANopen, mobile manipulation, motion control, calibration, robot bring-up; RealMan, PiPER, UR, Robotiq, Ranger Mini, RealSense, Livox.

AI & Software: Claude Code/Codex, Python, C/C++, PyTorch, OpenPI, LeRobot, VLA, flow matching, model training, evaluation and deployment; Isaac Sim, Genesis, Mujoco.

Engineering: Linux, SolidWorks, ANSYS, GPU infrastructure, supplier coordination, technical documentation.

English: TOEFL iBT 105, fluent in professional and academic English.